

Algebra 1 – Unit 3 Test C

Name: _____

Solving Equations

1. Solve for x .

$$x - 5 = 13$$

2. Solve for y .

$$2y + 9 = 25$$

3. **Word Problem:**

A small catering order included a \$6 delivery fee.

The total cost for five identical meals was \$41.

Write and solve an equation to find the price of each meal.

Multi-Step Equations

4. Solve for x .

$$4(x - 1) = 20$$

5. Solve for p .

$$3(p + 2) - 1 = 14$$

6. Solve for r .

$$7r - 4 = 3r + 12$$

7. Solve for m .

$$2(m + 5) = 3m + 1$$

8. Solve for t .

$$6(t - 4) + 3 = 6t - 5$$

Solving Inequalities

9. Solve and graph the solution on a number line.

$$4x - 1 < 11$$

10. Solve and graph the solution on a number line.

$$-4z \leq 8$$

Literal Equations

11. Solve for h .

$$V = \pi r^2 h$$

12. Solve for t .

$$N = 5h + 4t$$

13. Solve for x .

$$y = \frac{3}{4}x + b$$